Environment Canada
Environnement Canada
Service des pêches
et des sciences de la mer
DFO - Library / MPO - Bibliothèque
02004476

An Inventory of the Fisheries Resources of Ladysmith Harbour

Fisheries & Oceans Canada
Fisheries & Marine Service
Fisheries Management Regional Library
1090 W. Pender St., Vancouver, B.C.
Canada V6E 2P1

1975

by
J.H. Patterson

Internal Report Series No. PAC/I-75-1

Southern Operations Branch
Pacific Region

PAC
/I-75-1

An inventory of the
fisheries resources
Ladysmith Harbour

DEPARTMENT OF FISHERIES & OCEANS
FISHERIES & MARINE SERVICE REGIONAL LIBRARY
555 WEST HASTINGS STREET
VANCOUVER, B.C.
V6B 5G3

AN INVENTORY OF THE FISHERIES RESOURCES

OF LADYSMITH HARBOUR

Fisheries & Oceans Canada
Fisheries & Marine Service
Fisheries Management Regional Library
1090 W. Pender St., Vancouver, B.C.
Canada V6E 2P1

DFO LIBRARY
INSTITUTE OF OCEAN SCIENCES
PO BOX 6000
9860 WEST SAANICH RD
SIDNEY, BC CANADA V8L 4B2

BY

John H. Patterson

January 1975

Department of the Environment
Fisheries and Marine Service
Fisheries Operations South
Habitat Protection Unit

DEPARTMENT OF FISHERIES & OCEANS
FISHERIES MANAGEMENT REGIONAL LIBRARY
555 WEST HASTINGS STREET
VANCOUVER, B.C.
V6B 5G3

TABLE OF CONTENTS

	Page
LIST OF TABLES AND FIGURES	i
LIST OF PLATES	ii, iii
LIST OF APPENDICES	iv
ACKNOWLEDGEMENTS	v
INTRODUCTION	1
PROCEDURES	2
RESULTS AND DISCUSSION	3
SUMMARY	14
RECOMMENDATIONS	15
BIBLIOGRAPHY	16
APPENDICES	17

LIST OF TABLES AND FIGURES

		Page
TABLE 1	Herring Spawn Sites 1951-1974	11
FIGURE 1	Map of Ladysmith Harbour, sensitive fisheries habitats, and existing log handling facilities	4

LIST OF PLATES

		Page
PLATE 1	Raft booming of logs in front of Domans Industries sawmill (Ladysmith Forest Products).	47
2	Log booming grounds adjacent Ivy Green Park salt marsh	47
3	Bush Creek, 150' upstream from its mouth. Salt marsh vegetation bordering the stream course.	48
4	Bush Creek, downstream to mouth.	48
5	Bush Creek, associated salt marsh	49
6	Bush Creek, estuarine area.	49
7	Rocky Creek, stream-bed behind Saltair Logging Company property.	50
8	Rocky Creek, near mouth showing channel around Saltair Logging Company.	50
9	Ladysmith Harbour, immediately in front of Rocky Creek.	51
10	Salt marsh at Ivy Green Park, adjacent to mouth of Bush Creek.	52
11	Davis Lagoon, showing salt marsh at mouth of Stocking Lake Creek.	52
12	Estuarine area where Stocking Lake Creek flows into Stuart Channel.	53
13	Debris and loose logs from log handling facilities at head of Ladysmith Harbour.	53

LIST OF PLATES (Cont'd)

Photographs of substrate sample sites:		Page
PLATE 14	Lower intertidal area Bush Creek estuary	54
15	Mud flats, Davis Lagoon Mouth of Stocking Lake Creek	54
16	Lower intertidal region Holland Creek estuary	55
17	Salt Marsh (with debris) Holland Creek	55
18	Substrate sample site (S_5) Intertidal area (Fig.1)	56

LIST OF APPENDICES

		Page
APPENDIX A	Salt marsh definition	17
B	Substrate samples taken within Ladysmith Harbour	18
C	Herring spawn sites for those years when spawning occurred within Ladysmith Harbour	26
D	Photographs of intertidal mud flats - head of Ladysmith Harbour	43

ACKNOWLEDGEMENTS

The author wishes to thank M.J. Brownlee, K.W. Bond, and D. Moir, for their assistance with the format and technical aspects of this report.

Valuable information was provided by L.A. Webb, E.E. Epps, E.S. Robertson (Fisheries and Marine Service) and Don Outram (Biological Station, Nanaimo).

INTRODUCTION

Ladysmith Harbour, situated on the east coast of Vancouver Island, is an enclosed bay opening in a southeast direction into Stuart Channel.

Historically, Ladysmith Harbour has been one of the most productive shellfish areas on the east coast of Vancouver Island. Closures to the direct marketing of oysters found within Ladysmith Harbour, however, have been imposed since 1964 due to bacterial contamination.

Three sawmills are situated on the harbour shore, two near the Township of Ladysmith. The harbour provides protection for both intertidal and subtidal log storage.

Chum (Oncorhynchus keta) and coho (O. kisutch) salmon, as well as steelhead (Salmo gairdneri), and cutthroat (S. clarki clarki) trout, utilize streams flowing into the harbour. Prior to 1964, areas within the harbour were utilized as herring spawn sites. Subsequent reduction of herring spawn, and concern over further alienation of salmonid and herring rearing areas resulted in a survey of the fisheries resource by the Habitat Protection Unit, with the following objectives:

(1) to identify sensitive stream, estuary, and herring spawn sites within Ladysmith Harbour.

(2) to determine the impact of existing log handling and storage on (1).

(3) recommend protective guidelines or remedial measures to ensure productive viability of (1).

PROCEDURES

Ladysmith Harbour was visited by the writer from November 6, 1974 to November 8, 1974. The purpose of this trip was to survey various aspects of the fisheries (both salmon and herring) resource present in the area, and to establish the extent of harbour area presently utilized by log handling operations.

In order to survey one aspect of utilization of the harbour by rearing salmonids, it was necessary to assess visually the extent of vegetation associated with the three stream outlets (as well as peripheral areas throughout the harbour), and to take visual substrate evaluations at various sample sites.

Spawning records were examined to obtain a 10 year average of escapement for the following streams, all of which find outlet along the southern side of Ladysmith Harbour:

1. Bush Creek
2. Holland Creek
3. Rocky Creek, and
4. Stocking Lake Creek.

Herring spawn records for the past twenty-three years were examined in order to ascertain the historical utilization of the interior reaches of the harbour as spawning grounds. These results were compared with more recent herring spawn distribution.

Visual estimates were made as to the extent and type of log handling practices within Ladysmith Harbour.

Assessment of the shellfish industry within the harbour was impossible due to time limitation and tidal conditions. Data pertaining to shellfish areas were derived from land use maps from the Department of Land, Forests and Water Resources.

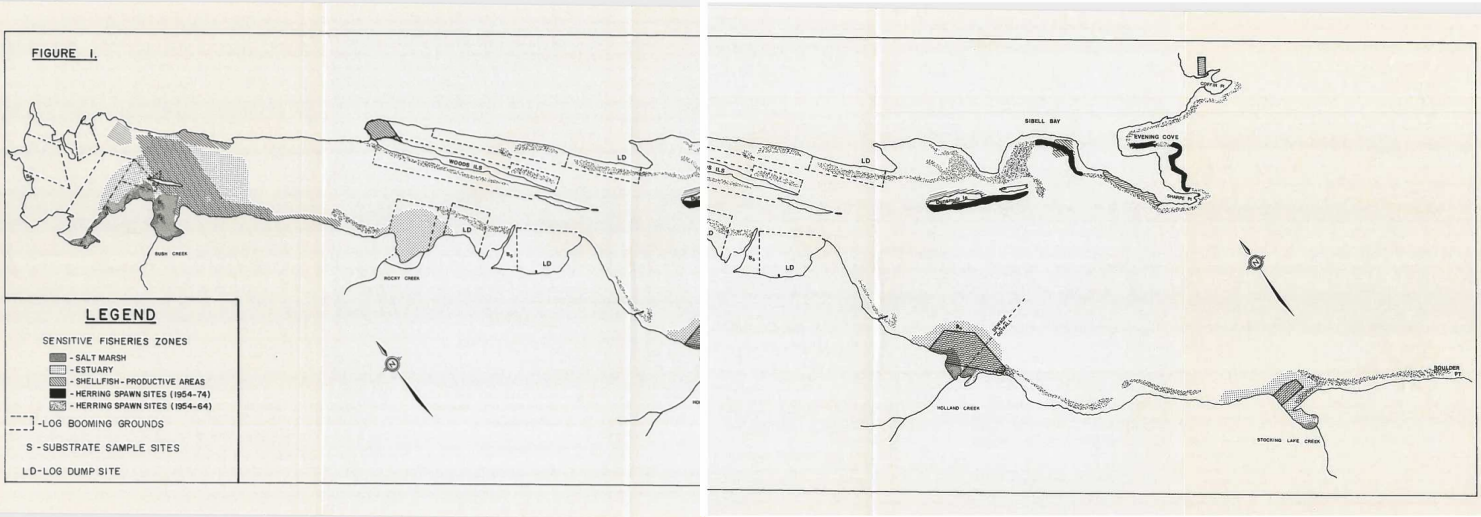
RESULTS AND DISCUSSION

1. Log Handling

The log handling facilities and extent of present areas under raft type booms (approx. scale) are shown on Figure 1. (Plate 1). Log dumping is of the A-frame type, with the logs being vertically hoisted from trucks or train cars, and either rolled or dumped into the harbour. Log booming grounds in intertidal areas at the head of the harbour exist above substrate consisting mainly of mud and clay (Plate 2). However, it is difficult to estimate the extent of algae growth in these areas prior to the issuance of log booming leases. A survey of the literature indicates that log booming and subsequent bark slouphage impose stress upon the harbour environment in the following ways: (Dr Dan B. Quayle from a letter of October 28, 1971, from K. Radway Allan, Biological Station, Nanaimo, to Mr J. WHITLAW of Duncan, B.C.)[1]

a) destruction of benthic organisms (development of noxious gas)

FIGURE I.



LEGEND

SENSITIVE FISHERIES ZONES

- SALT MARSH
- ESTUARY
- SHELLFISH - PRODUCTIVE AREAS
- HERRING SPAWN SITES (1954-74)
- HERRING SPAWN SITES (1954-64)

-LOG BOOMING GROUNDS

S -SUBSTRATE SAMPLE SITES

LD-LOG DUMP SITE

b) reduction of sunlight and the subsequent growth of algae utilized by spawning herring and rearing salmonids.

c) possible disruption of normal two-layered circulation system which may lead to alteration of normal temperature and salinity regimes.

A further study[4] indicated that increased bark deposits in areas of log dumping increase oxygen uptake for the sediments at these sites. This increased oxygen demand may depress oxygen levels in the overlying water "below levels necessary for the maintenance of a productive biological community"[4]. Bottom deposits prevent an effective barrier to the recolonization of the area to benthic life, even after log dumping operations have ceased.

2. Substrate and Aquatic Vegetation Survey

Visual estimates of substrate and vegetation were made at seven sites within the harbour. A percent estimate of substrate composition was made following the estuary inventory program format (Appendix B). In general, the substrates at the surveyed areas were composed of 70 to 90% silt clay to gravel 1/4-2" (silted) range. Substrate site S_7 typifies the northeast side of Ladysmith Harbour. Exposed bedrock with steep dropoffs and steep shoreline predominated. Larger size gravel was present. Oysters and other species of shellfish were present in these types of areas.

Vegetation at all substrate sites was severely limited (Plates 14-18). Mud flats or exposed bedrock characterized most intertidal areas. Timing and heights of low tides made lower intertidal investigations difficult. No rockweed areas or eelgrass beds were observed, while only isolated areas of fucus spp. were noted.

3. Escapement to Major Streams (10 yr. mean)

A) BUSH CREEK - flowing into Ladysmith Harbour at Ivy Green Park 2 miles north of Ladysmith (Plates 3, 4, 5 and 6). Spawning occurs in the lower 1.5 miles being restricted by an impassable obstruction (rock falls).

	Chum	Coho
1970	6700	—
1973	2650	150
10 yr. mean	3300	150

B) HOLLAND CREEK - flows into Ladysmith Harbour 2.75 miles south of Williams Point. Spawning occurs for the first 2 miles above tidal influence. A natural rock falls at this point is impassable to fish.

	Chum	Coho	Steelhead
1973	3700	60	
10 yr. mean	5580	74	observed

C) ROCKY CREEK - flows into the harbour just above Williams Point near Saltair Logging Company Sawmill site. No escapement records exist for this stream, but fisheries officer E. Robertson indicates that, historically, chum

and coho did utilize the stream. At present, the escapement is nil, or at least unobserved. A cement culvert under Highway #19 and a rock falls immediately downstream may impose an obstacle to upstream migration. The original water course has been diverted around the Saltair Logging Company property (Figure 1). Spawning gravel exists in the lower reaches. However, the estuarine area (41 acres) is heavily utilized for log storage (Plates 7, 8, 9). A salt marsh or associated estuarine vegetation was not observed.

D) STOCKING LAKE CREEK - flows into Stuart Channel. A natural obstruction and a cement dam exist 500 yards upstream from Davis Lagoon.

	Chum	Coho
1973	2200	20
10 yr. mean	3000	30

4. Salmonid Rearing Areas

Bush Creek Estuary (Plates 3, 4, 5 and 6) consists of a large area approximately 140 acres comprised mainly of mud flats under tidal influence. Vegetation was not observed in the lower intertidal zone (Substrate S_2). The middle intertidal zone is comprised principally of sand and gravel 2-4" (S_1) and some fucus spp. were noted. The upper to supra tidal region consists of an extensive salt marsh (600' long by 300' wide). Salicornia virginica and salt grass (Distichlis picata) predominate. This salt marsh

extends for approximately 2,000' around the point adjacent to Ivy Green Park (Plates 2, 10). The same two species of plant are present. Tidal drift of insects from the marsh along with copepods and isopods observed in the lower marsh regions would constitute one of the principal sources of food for rearing salmonids.

The extensive log booming in this area is confined to the lower intertidal mud flats. Evidence of erosion of this salt marsh was observed in areas adjacent to log booms. Retardation of increased salt marsh growth and encroachment of existing areas may result from log booms grounded during low tides, causing tidal rivulets and slouphage.

Salt marsh communities exist in the delta at the mouth of Holland Creek (60 acres). The lower intertidal area (substrate S_4) is comprised of mud flats of approximately 12 acres. Fucus spp. were noted, as well as localized pockets of Percusaria percursa. Salicornia virginica, with other salt marsh vegetation, is abundant in the upper intertidal zone. Tidal exportation of salt marsh invertebrates, as well as invertebrate drift from Holland Creek, may constitute a significant portion of the diet for rearing salmon.

The third area examined, as to the rearing potential, was the estuarine flats at the mouth of Stocking Lake Creek (30 acres). Vegetated substrate is restricted to the upper

25-35% of Davis Lagoon (Plate 11). The remaining area of the lagoon consists primarily of mud flats with virtually no vegetation. Stocking Lake Creek flows through this lagoon, under the Chemainus Road Bridge into Stuart Channel (Plate 12). This outlet was examined at low tide for vegetation present and for rearing potential. Salt marsh vegetation and lower intertidal algae were not observed. The substrate consisted mainly of mud and fine silt. A large proportion of nearshore rearing of fry emerging from Stocking Lake Creek would be derived from the upper Davis Lagoon salt marsh.

Information pertaining to the contribution of marine zooplankton or other potential food organisms within Ladysmith Harbour to the rearing population of these three systems, was not collected.

5. Shellfish

The status of shellfish production and oyster leasing (Figure 1) within Ladysmith Harbour was not studied in detail during this survey.

Closures to direct marketing of oysters were imposed within the harbour following sanitary and bacteriological surveys in 1962, 1963 and 1964.[2] This contamination of oyster growing waters was mainly the result of the Town of Ladysmith's discharge of raw sewage into the inner harbour.

A bacteriological survey carried out by E.P.S. of Environment Canada indicates that bacterial contamination of the inner harbour waters (division line is the constriction between Slag Pt. and Woods Island) has decreased since 1962. However, this report[2] also suggests the closure of 40 acres of "restricted" oyster leases in the Holland Bank area, since the sewage outfall from the primary treatment facilities of the Town of Ladysmith is situated off this Bank. The report cited states that with certain remedial measures pertaining to sewage outfall conversion and other technical aspects, "120 acres of the existing restricted oyster leases in the inner harbour could be approved for direct marketing". Productive shellfish areas described in Figure 1 were taken from land use maps made available from the Department of Lands, Forests and Water Resources.

6. Herring Spawning Areas

Information on the distribution of herring spawn within and about Ladysmith Harbour was compiled in an attempt to compare historic and recent distribution (Table 1; Figure 1). Data obtained from Don Outram (Biological Station, Nanaimo) and L.A. Webb (Fisheries and Marine Service) indicate a reduction of the area utilized by spawning herring inside the harbour, over the last 23 years.

Table 1 indicates that herring have not utilized the inner harbour spawning areas (particularly from the

TABLE 1

YEAR	HERRING SPAWN SITES	
1951	*Ladysmith Harbour (No particular location given)	
	*Shell Beach	
	Dunsmuir Island	
1952	*Ladysmith Harbour	
1953	*Government Wharf	
	*Shell Beach	
	Dunsmuir Island	
1954	*Government Wharf	
	Boulder Point	
	Dunsmuir Island	
1955	*Ladysmith Harbour	
1956	*Ladysmith Harbour	
	Boulder Point	
	Dunsmuir Island	
1957	*Ladysmith Harbour	
1958	*Ladysmith Harbour	
	*Govt. wharf up harbour opposite Woods Island	eelgrass
1960	same as 1958	eelgrass/rockweed
	Sharpe Point	japweed/rockweed
	*Davis Lagoon to Govt. wharf	
	Sibell Bay	
	Davis Lagoon to Boulder Point	eelgrass/rockweed
1961	*Govt. wharf up Harbour	sparse eelgrass
1962	Dunsmuir Island	
	No herring spawn within harbour	
1970	Dunsmuir Island	
	Sharpe Point	
1971	Dunsmuir Island	
	Sharpe Point	
1973	NIL within Harbour	
1974	Evening Cove (Sharpe Pt.)	

* Within Ladysmith Harbour

Government Wharf to the head of the harbour) since 1964. Ladysmith Harbour has been given as a general location for herring spawn between 1951 and 1958, but not between 1958 and 1974. Dunsmuir Island has been utilized consistently from 1951 to 1971, but not in the last 3 years. As shown on Figure 1, the southern shoreline of the harbour was utilized as spawning sites at various different years up to 1964. No herring spawn occurred on this side of the harbour from Boulder Point to the Standard Oil marina between 1964 and 1974.

It is quite clear that there has been a reduction in the areas utilized by herring as spawning grounds, within Ladysmith Harbour. The only regions that have consistently been used by herring over the past twenty years are Dunsmuir Island, Sibell Bay, Sharpe Point and Evening Cove, the latter two areas falling outside of the harbour.

There appears to be a trend through the east coast of Vancouver Island towards spawning in more open, exposed grounds (pers. comm., T. Haegle, Biological Station, Nanaimo). The reduction of the extent of herring spawn within the harbour may reflect this trend toward spawning in more exposed areas or may be the result of an elimination of preferred spawning sites. Some of these sites (particularly those sites between Dunsmuir Island and the northern tip of Woods Island) are coincidental with log storage.

Personal communication with Don Outram (Biological Station, Nanaimo) has revealed that reduction of eelgrass, rockweed and japweed beds within the harbour may be one of the major contributing factors in the failure of herring to spawn within Ladysmith Harbour (inside Dunsmuir Island). Decreased light penetration associated with log booming facilities, may have contributed to this reduction.

SUMMARY

1. Sensitive fisheries zones identified within Ladysmith Harbour include stream estuaries, herring spawn sites, salt marshes and shellfish areas.

2. Annual salmon escapements (10 year mean) to the three major producing streams (Bush Creek, Holland Creek, Stocking Lake Creek) total 11,880 chum and 260 coho.

3. Herring spawn distribution (Figure 1) has decreased markedly since 1964 within the harbour limits. Sites presently utilized include Dunsmuir Island, Sharpe Point and Coffin Point.

4. Vegetation below the supratidal fringe is restricted generally to fucus spp. and salt marsh communities (Appendix A). Salt marsh communities are associated with the three major salmon-producing streams and the foreshore of Ivy Green Provincial Park.

5. Intertidal log storage where coincidental with salt marsh communities appears to accelerate erosion of these sites.

6. Extensive log booming and storage occurs within Ladysmith Harbour. Water based log sorting and A-frame dump facilities are present on both sides of the harbour.

7. Existing log handling operations coincidental with identified sensitive fisheries zones include:

   (a) intertidal log storage adjacent to Ivy Green Park

   (b) Rocky Creek Estuary

   (c) Historical utilized herring spawn sites (Figure 1)

   (d) Lower, middle intertidal zones at Bush Creek estuary.

8. Respective estuary acreage and acreage presently utilized for log handling and storage.

	Estuary Acreage	Acreage Utilized
Bush Creek	140	0*
Holland Creek	60	0
Rocky Creek	41	41
Stocking Lake Creek	30	0

* Periodic log transport observed.

RECOMMENDATIONS

To maintain a viable fisheries resource within Ladysmith Harbour, it is recommended that:

1. Log storage and/or dumping facilities not be permitted within, or immediately adjacent to, those areas delineated as fisheries sensitive zones (Figure 1).

2. Existing log storage be phased out adjacent to the following zones:

   (a) Salt marsh adjacent to Ivy Green Park

   (b) Bush Creek lower intertidal estuarine zone

   If deemed necessary by Fisheries Personnel, further remedial measures be implemented on completion of the above.

3. Should further studies indicate the possibility of restoring historically utilized herring spawn sites within Ladysmith Harbour and the salmon escapement to Rocky Creek, existing industrial activities at identified sites, be phased out and remedial measures implemented.

4. Further studies be implemented to determine:

   (a) potential for salmon enhancement to Bush, Holland and Stocking Lake Creeks.

   (b) distribution and food utilization of rearing salmonids within Ladysmith Harbour.

5. Early discussions be undertaken between representatives of the appropriate Industries and resource agencies to develop specific protective procedures and formalize phase-out sequences.

BIBLIOGRAPHY

1. McALLISTER, C.D. Phytoplankton and Physical Conditions in Ladysmith Harbour. B.C. Department of Fisheries Annual Report, 1956.

2. TEVENDALE, T.J. Shellfish growing water sanitary survey of Ladysmith Harbour, British Columbia, 1970. D.O.E. Environmental Protection Service, Vancouver, B.C. Report Number EPS 5-WP-73-1.

3. KOZLOFF, E.M. Seashore Life of Puget Sound, the Strait of Georgia, and the San Juan Archipelago. J.J. Douglas Ltd., Vancouver, 1973.

4. Log Storage and Rafting in Public Waters. A Task Force Report approved by Pacific North-west Pollution Control Council, August, 1971.

APPENDIX A

Salt Marsh definition

The uppermost region of the estuary, immediately below the supratidal area, has been described as a salt marsh. The following are some typical characteristics of a salt marsh:[3]

1. generally bordering bays and estuaries,
2. inundated only by the highest tides,
3. typically rather flat, except for shallow pools and narrow channels,
4. characterized by the presence of a flowering plant, Salicornia virginica,
5. a prevalent form of true grass, salt grass Distichlis picata
6. associated fauna:

   2 small snails Phytia myosotis and Assiminea californica

   Littorina sitkana

   Hemigraphsus nudus

   Orchestia braskiane, amphipod

APPENDIX B

SUBSTRATE SAMPLES TAKEN WITHIN LADYSMITH HARBOUR

(see Figure 1 for locations)

S-1

ESTUARY INVENTORY PROGRAM

BUSH CREEK - middle intertidal

200' deep

(1) Intertidal area in acres

(2) Vegetated intertidal area — extent in %

(3) Species (flora and fauna) some fucus — extent in % or abundance

(4) Supra-tidal cover (high water to level ground)

a)	Vegetation	extent in %
b)	Pavement	
c)	Buildings	
d)	Bare soil	

(5) Substrate

		Code	
a)	Bedrock	1	extent in %
b)	Boulder/cobble	2	
c)	Gravel 4-6" (clean)	3	
d)	Gravel 4-6" (silted)	4	
e)	Gravel 2-4" (clean)	5	30%
f)	Gravel 2-4" (silted)	6	
g)	Gravel ¾-2" (clean)	7	
h)	Gravel ¾-2" (silted)	8	10%
i)	Sand	9	40%
j)	silt/mud/clay	10	20%
k)	organic	11	

(6) Log booming and storage

a)	Inter-tidal	extent in acres.
b)	Sub-tidal	

ESTUARY INVENTORY PROGRAM S-2

BUSH CREEK — Lower intertidal From Island up to shore for 300'

(1) **Intertidal area in acres**

(2) **Vegetated intertidal area** LOWER NIL **extent in %**

(3) **Species (flora and fauna)** **extent in % or abundance**

(4) **Supra-tidal cover (high water to level ground)**

a) **Vegetation** Marsh Flats **extent in %** Extensive
b) **Pavement** (Saltmarsh)
c) **Buildings**
d) **Bare soil**

(5) **Substrate**

	Code	extent in %
a) Bedrock	1	
b) Boulder/cobble	2	
c) Gravel 4-6" (clean)	3	
d) Gravel 4-6" (silted)	4	
e) Gravel 2-4" (clean)	5	
f) Gravel 2-4" (silted)	6	
g) Gravel ¼-2" (clean)	7	5%
h) Gravel ¼-2" (silted)	8	
i) Sand	9	15%
j) silt/mud/clay	10	80%
k) organic	11	

(6) **Log booming and storage**

a) **Inter-tidal** **extent in acres.**
b) **Sub-tidal**

ESTUARY INVENTORY PROGRAM S-3

AREA OF HARBOUR
ADJACENT MOUTH OF BUSH
CREEK

(1) Intertidal area in acres

(2) Vegetated intertidal area — extent in %

(3) Species (flora and fauna) — extent in % or abundance

(4) Supra-tidal cover (high water to level ground)

a) Vegetation	90%	extent in %
b) Pavement		
c) Buildings		
d) Bare soil	10%	

(5) Substrate

	Code	
a) Bedrock	1	extent in %
b) Boulder/cobble	2	
c) Gravel 4-6" (clean)	3	
d) Gravel 4-6" (silted)	4	
e) Gravel 2-4" (clean)	5	
f) Gravel 2-4" (silted)	6	40%
g) Gravel ¼-2" (clean)	7	
h) Gravel ¼-2" (silted)	8	50%
i) Sand	9	
j) silt/mud/clay	10	10%
k) organic	11	

(6) Log booming and storage

a) Inter-tidal	NIL	extent in acres.
b) Sub-tidal	NIL	

No Visible Debris

Some Large Pieces Wood
Small Logs

Many Chum Present

ESTUARY INVENTORY PROGRAM S-4

HOLLAND CREEK
Middle Intertidal area

(1) Intertidal area in acres

(2) Vegetated intertidal area — extent in %

(3) Species (flora and fauna) — extent in % or abundance

(4) Supra-tidal cover (high water to level ground)
- a) Vegetation — extent in %
- b) Pavement
- c) Buildings
- d) Bare soil

(5) Substrate

	Code	extent in %
a) Bedrock	1	
b) Boulder/cobble	2	
c) Gravel 4-6" (clean)	3	
d) Gravel 4-6" (silted)	4	
e) Gravel 2-4" (clean)	5	20%
f) Gravel 2-4" (silted)	6	
g) Gravel ¼-2" (clean)	7	
h) Gravel ¼-2" (silted)	8	50%
i) Sand	9	20%
j) silt/mud/clay	10	10%
k) organic	11	

(6) Log booming and storage
- a) Inter-tidal — extent in acres.
- b) Sub-tidal

Boulder/cobble underneath

ESTUARY INVENTORY PROGRAM

S-5
Between Domans Log Dump & Pacific Log Dump
Area #4
Supra-Tidal to Upper Middle Intertidal

(1) Intertidal area in acres

(2) Vegetated intertidal area NIL extent in %

(3) Species (flora and fauna) extent in % or abundance

(4) Supra-tidal cover (high water to level ground)

a) Vegetation NIL extent in %
b) Pavement Boat Launch
c) Buildings Govt. Wharf
d) Bare soil

(5) Substrate

	Code	extent in %	
a) Bedrock	1		
b) Boulder/cobble	2		
c) Gravel 4-6" (clean)	3		
d) Gravel 4-6" (silted)	4		
e) Gravel 2-4" (clean)	5		
f) Gravel 2-4" (silted)	6	10%	
g) Gravel ¼-2" (clean)	7	30%	
h) Gravel ¼-2" (silted)	8		▼ 20%
i) Sand Coarse to Grav.	9	60%	▼ 60%
j) silt/mud/clay	10		▼ 20%
k) organic	11		

(6) Log booming and storage

a) Inter-tidal extent in acres.
b) Sub-tidal

▼ Upper to middle intertidal
Bark Chips
Old Oyster Shells
No Vegetation
Few Large Pcs. Wood

ESTUARY INVENTORY PROGRAM

ADJACENT SAWMILL AT HEAD OF HARBOUR

(1) Intertidal area in acres

(2) Vegetated intertidal area — extent in %

(3) Species (flora and fauna) — extent in % or abundance

(4) Supra-tidal cover (high water to level ground)

- a) Vegetation — extent in %
- b) Pavement
- c) Buildings
- d) Bare soil

(5) Substrate

		Code	
a)	Bedrock	1	extent in %
b)	Boulder/cobble	2	
c)	Gravel 4-6" (clean)	3	
d)	Gravel 4-6" (silted)	4	
e)	Gravel 2-4" (clean)	5	
f)	Gravel 2-4" (silted)	6	
g)	Gravel ¼-2" (clean)	7	
h)	Gravel ¼-2" (silted)	8	10%
i)	Sand	9	
j)	silt/mud/clay	10	90%
k)	organic	11	

(6) Log booming and storage

- a) Inter-tidal — extent in acres.
- b) Sub-tidal

Extensive Floating Debris
Many Large Loose Logs
Little Vegetation

ESTUARY INVENTORY PROGRAM

S-7
EAST SIDE

(1) Intertidal area in acres

(2) Vegetated intertidal area — extent in %

(3) Species (flora and fauna) — extent in % or abundance

(4) Supra-tidal cover (high water to level ground)

a) Vegetation	extent in %	
b) Pavement	No Vegetation	
c) Buildings	Some Marginal	
d) Bare soil	Sabeornia	

(5) Substrate

	Code	extent in %
a) Bedrock	1	10%
b) Boulder/cobble	2	
c) Gravel 4-6" (clean)	3	
d) Gravel 4-6" (silted)	4	
e) Gravel 2-4" (clean)	5	
f) Gravel 2-4" (silted)	6	45%
g) Gravel ¼-2" (clean)	7	
h) Gravel ¼-2" (silted)	8	
i) Sand	9	35%
j) silt/mud/clay	10	10%
k) organic	11	

(6) Log booming and storage

a) Inter-tidal — extent in acres.

b) Sub-tidal

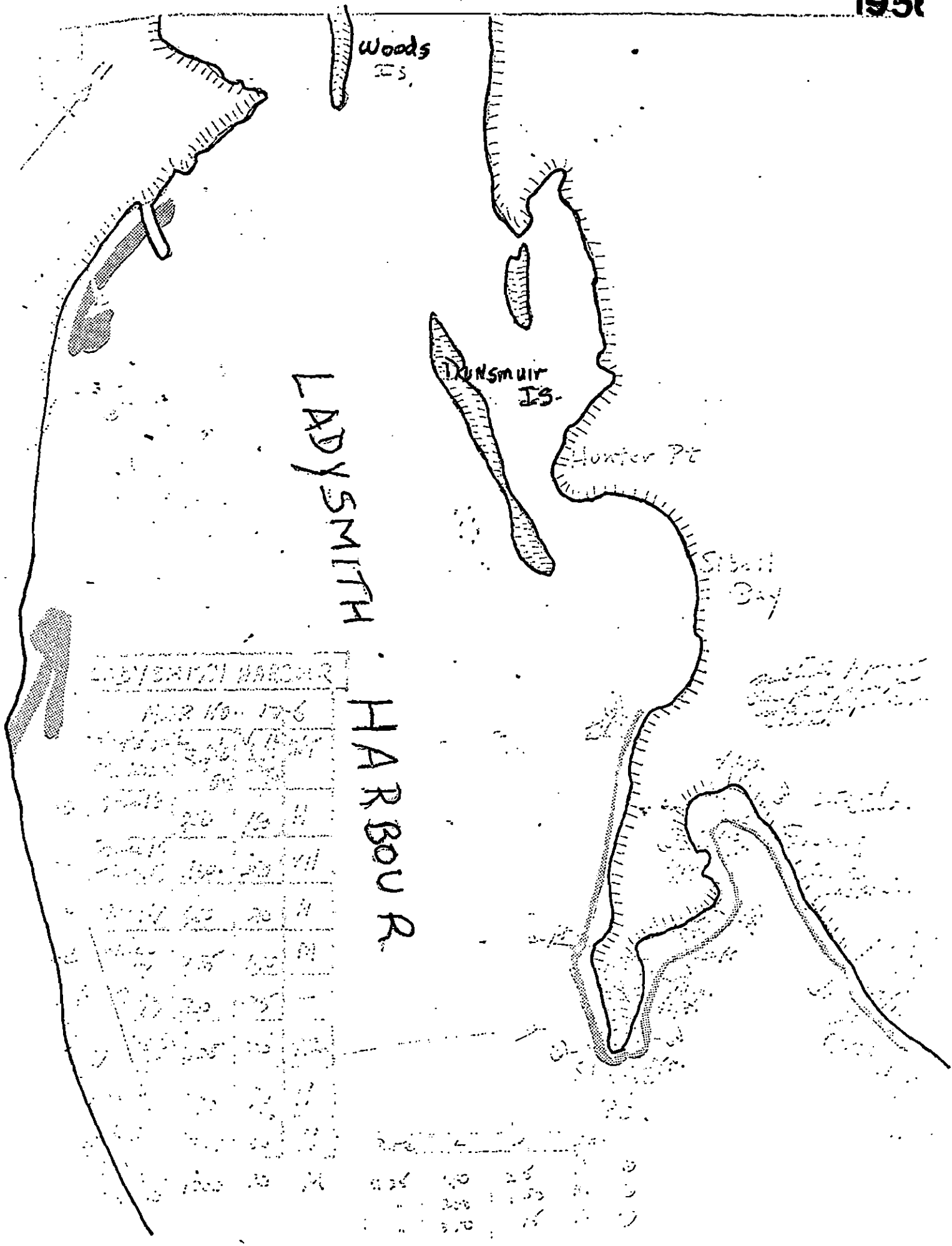
Oysters quite abundant (very silted)

Logs on beach
debris (bark)

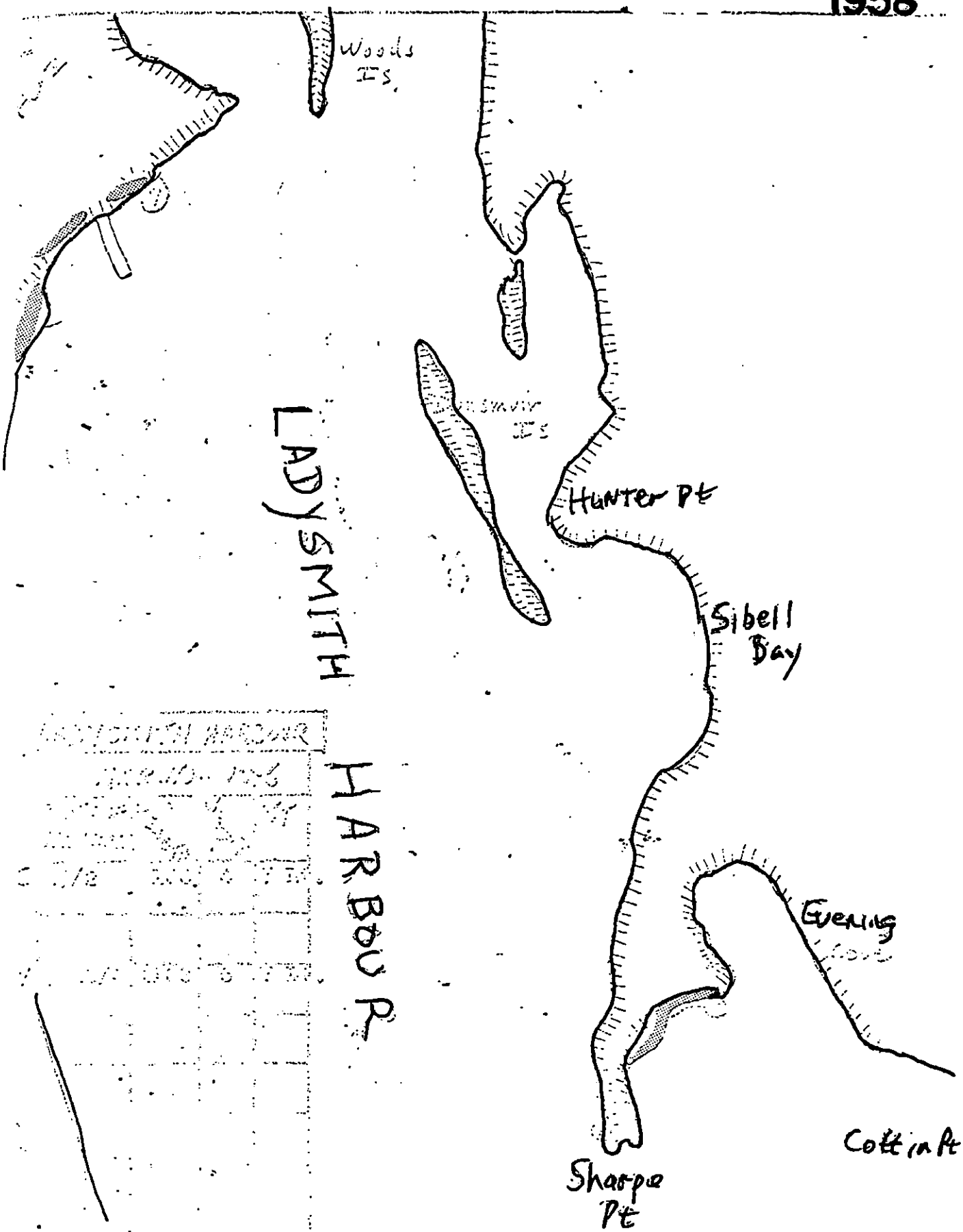
APPENDIX C

HERRING SPAWN MAPS* FOR THOSE YEARS WHEN SPAWNING OCCURRED WITHIN LADYSMITH HARBOUR

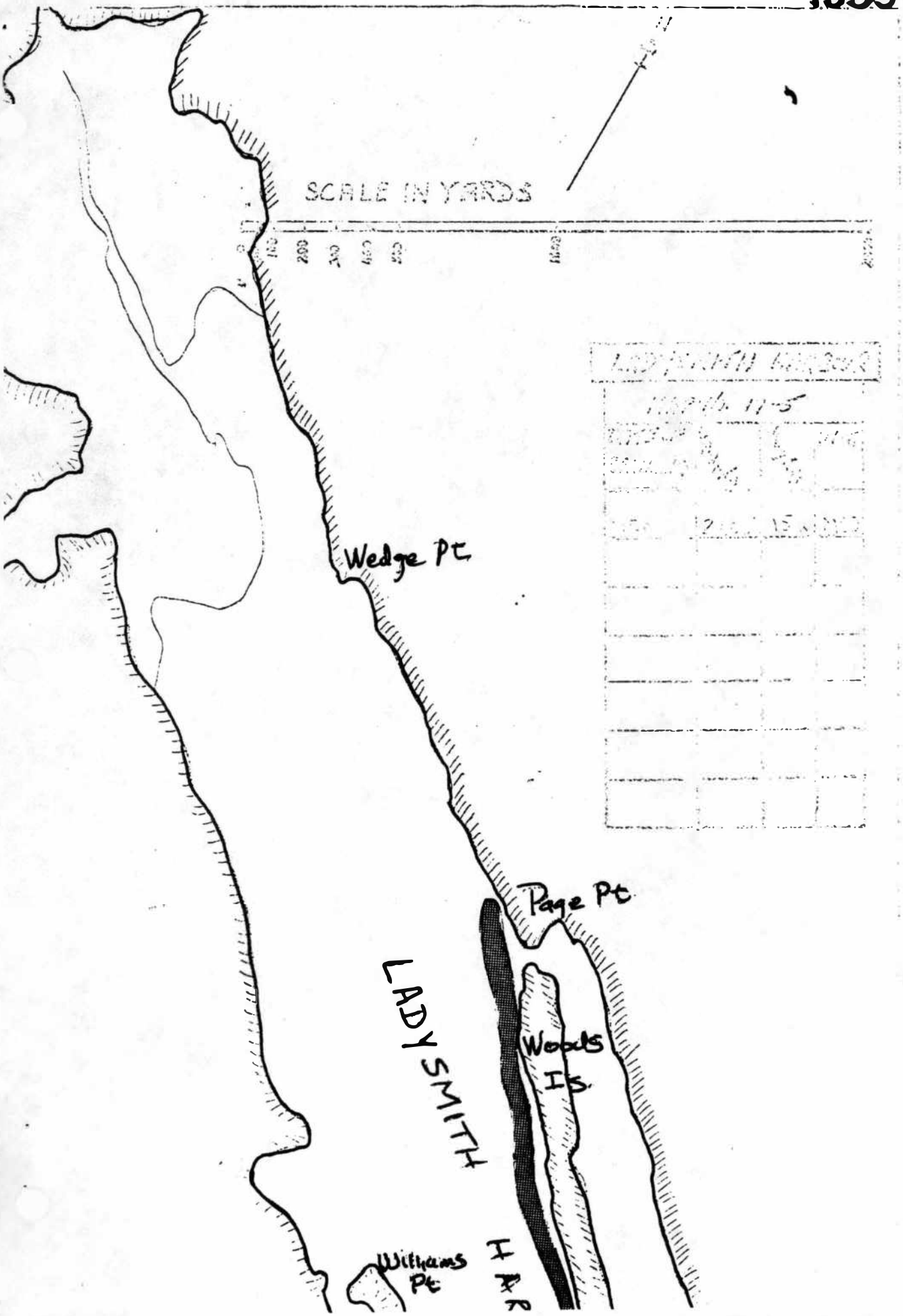
* COPIES OF FISHERIES OFFICERS' FIELD MAPS


Woods
Is.
Trunsmuir
Is.
Hunter Pt
LADYSMITH HARBOUR

1958


Woods Is.
LADYSMITH HARBOUR
Hunter Pt
Sibell Bay
Evening Cove
Cottin Pt
Sharpe Pt

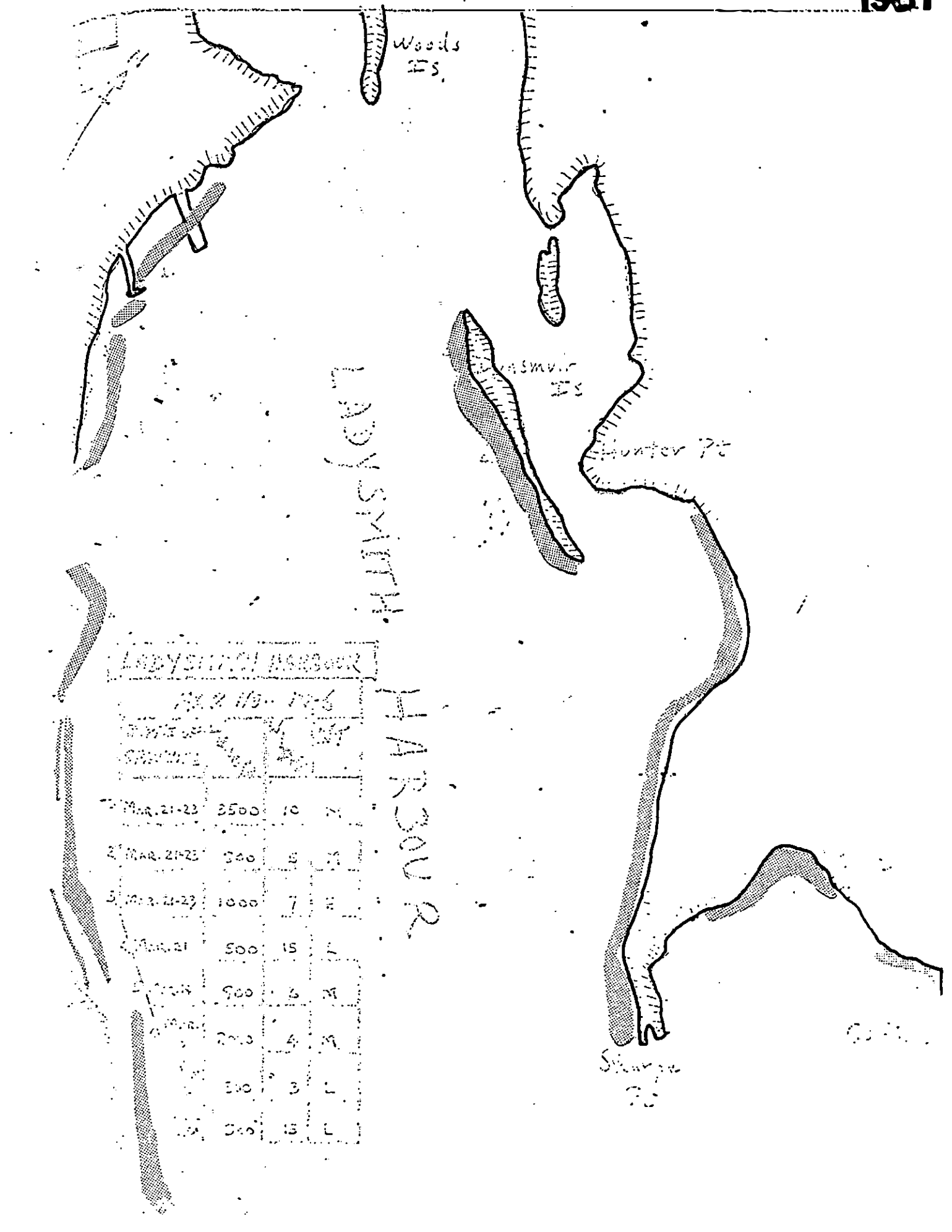
1959

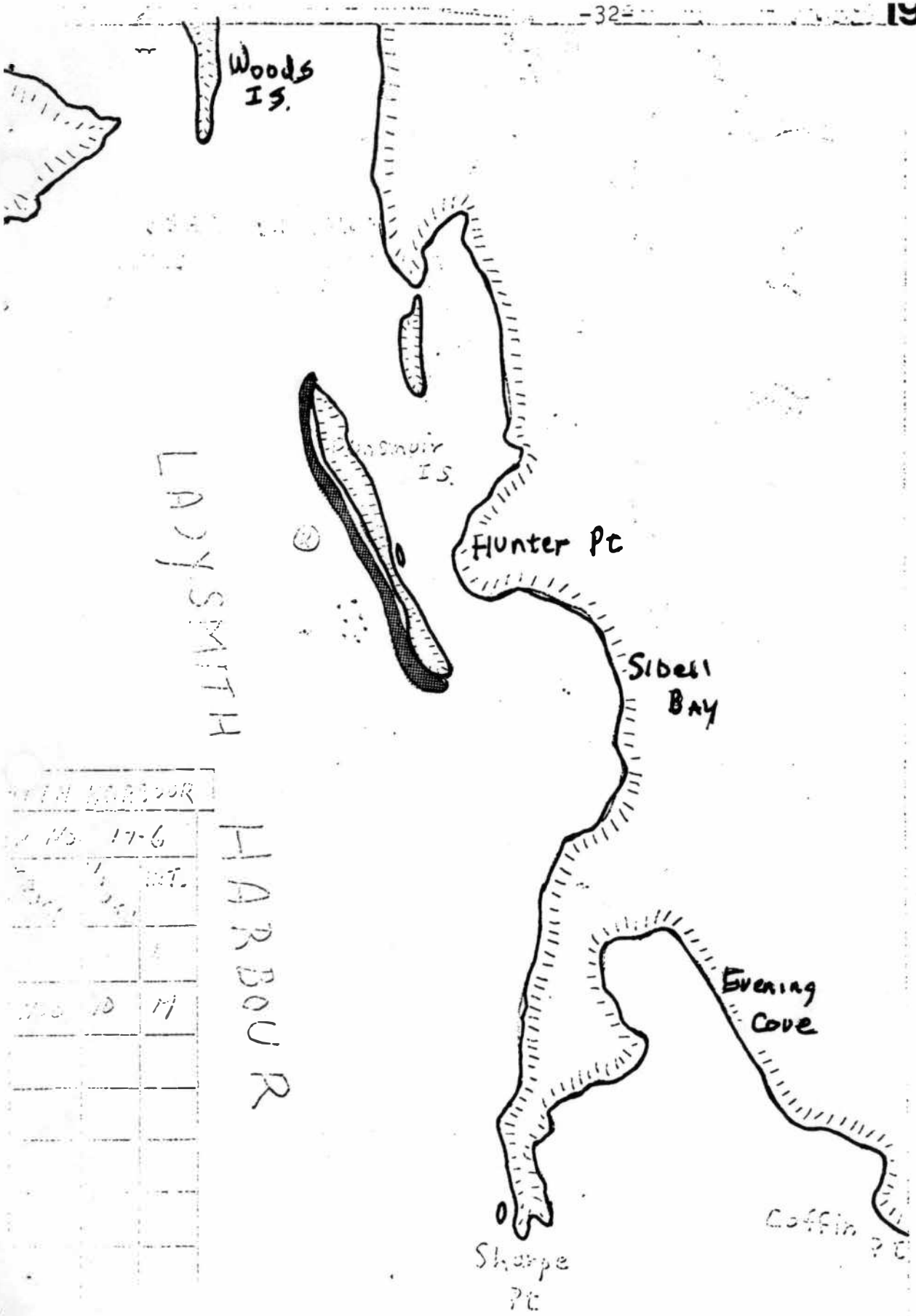

SCALE IN YARDS
Wedge Pt
Page Pt
LADYSMITH
Woods Is
Williams Pt

1959

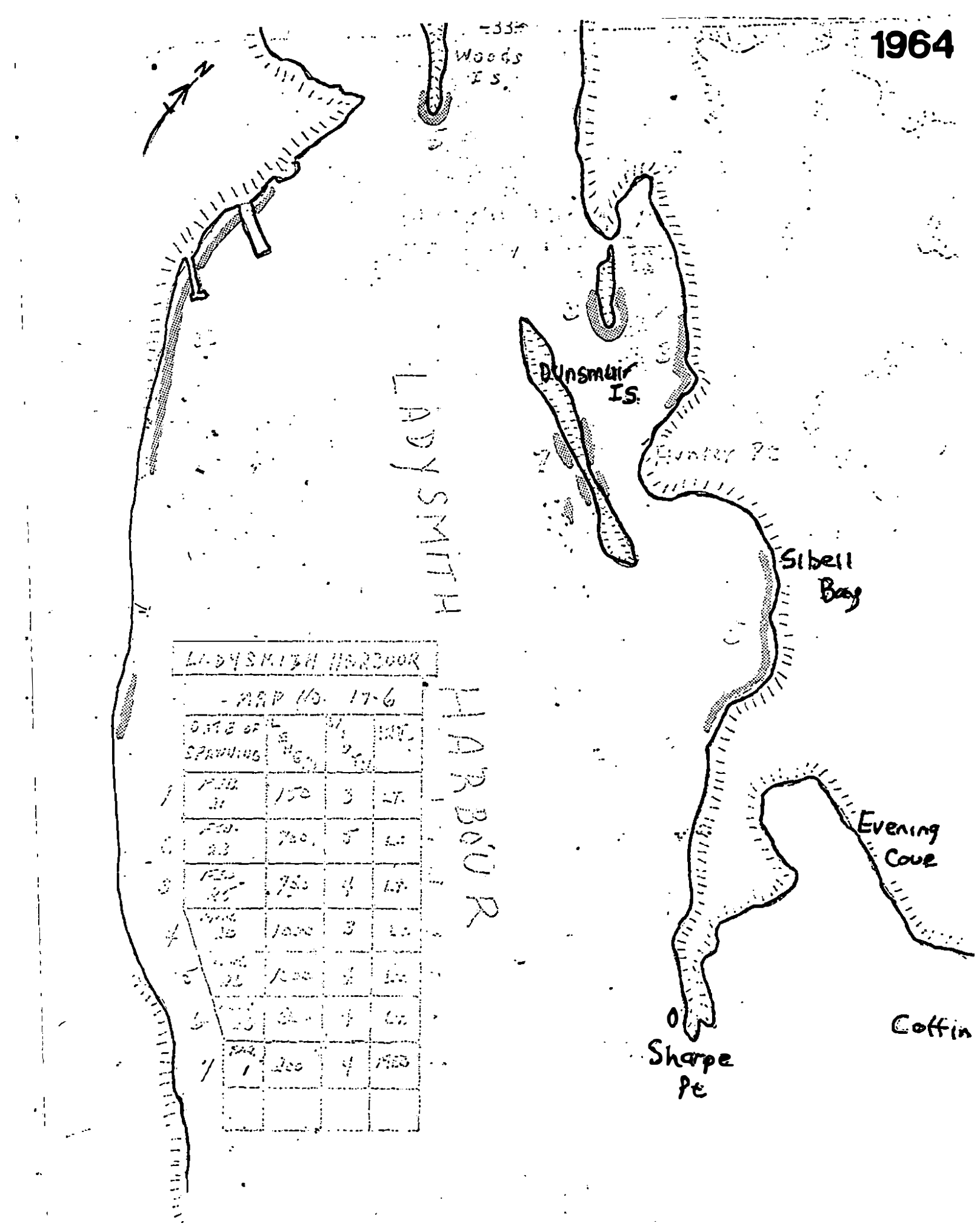
Woods
Is.
Hunter Pt
Sibell
Bay
Evening
Cove
LADYSMITH HARBOUR

1961




Woods Is.
Hunter Pt
Sidell Bay
Evening Cove
Sharpe Pt
Coffin Pt
LAUKSMITH HARBOUR

1964

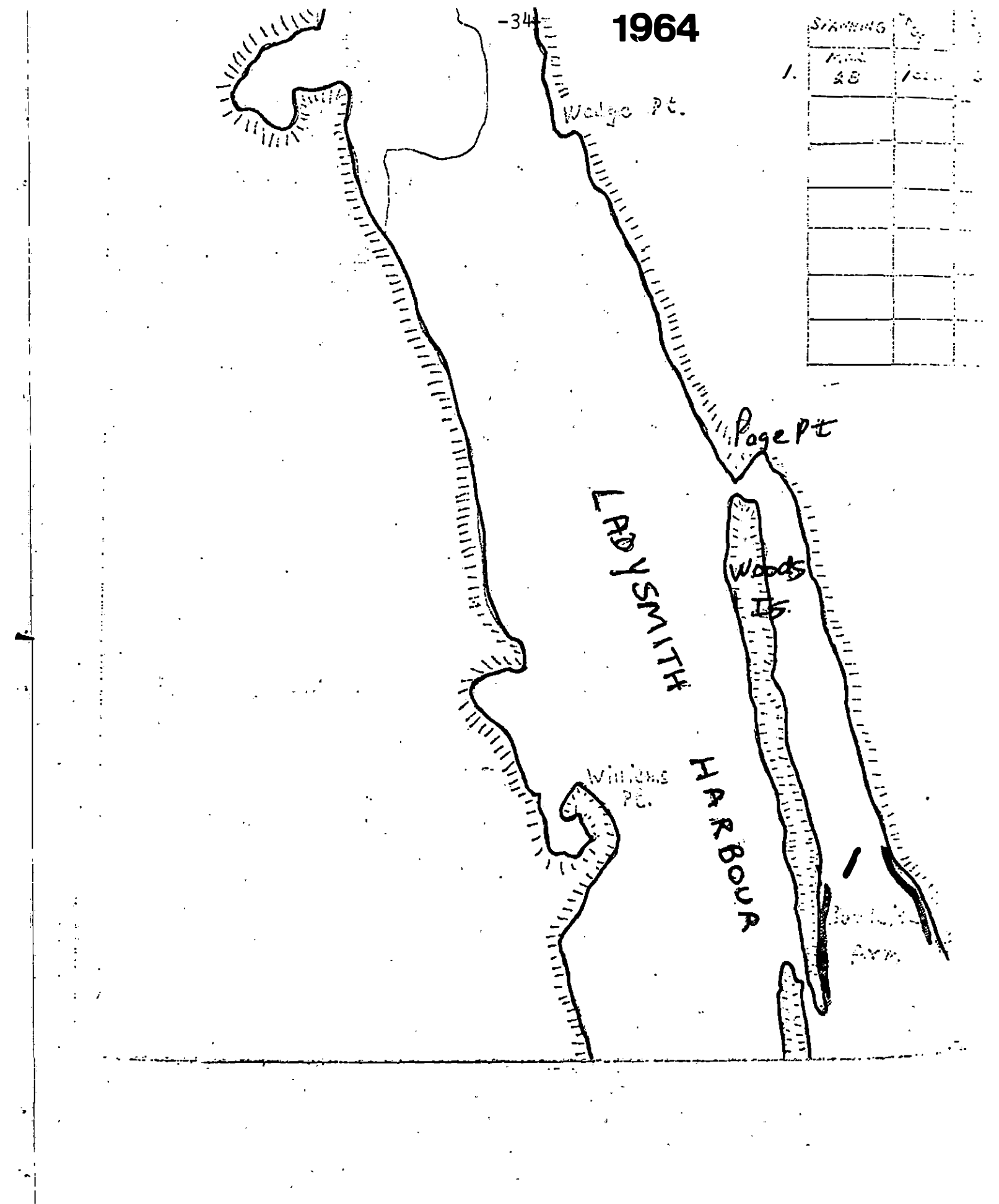


LADYSMITH HARBOUR

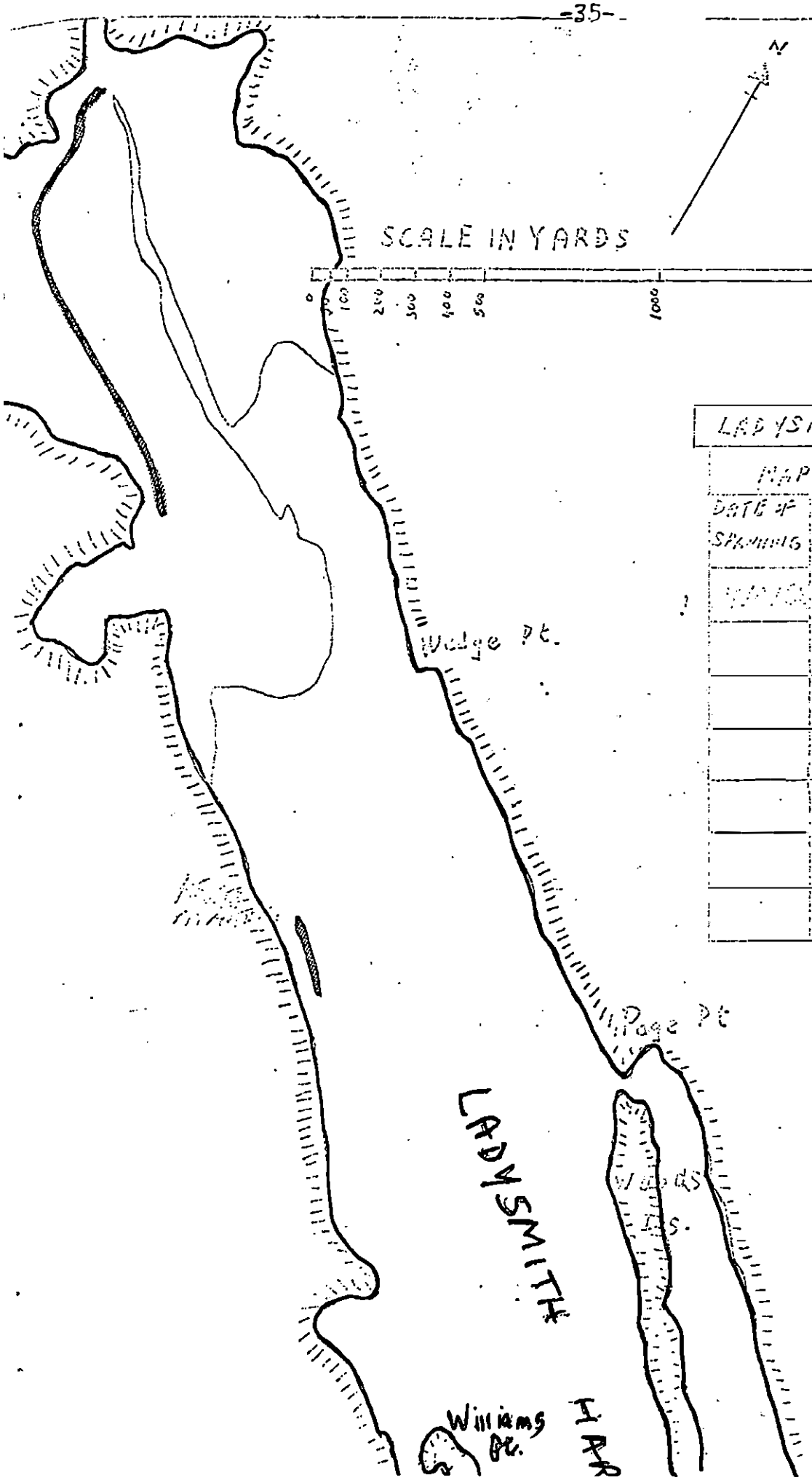
- MAP NO. 17-6

	DATE OF SPAWNING	[illegible]	[illegible]	[illegible]
1	[illegible] 31	150	3	[illegible]
[illegible]	[illegible] 23	750	5	[illegible]
3	[illegible] 24	750	4	[illegible]
[illegible]	[illegible] 25	1000	3	[illegible]
[illegible]	[illegible] 22	1250	5	[illegible]
[illegible]	[illegible] 20	500	4	[illegible]
1	MAR 1	200	4	MED

1964



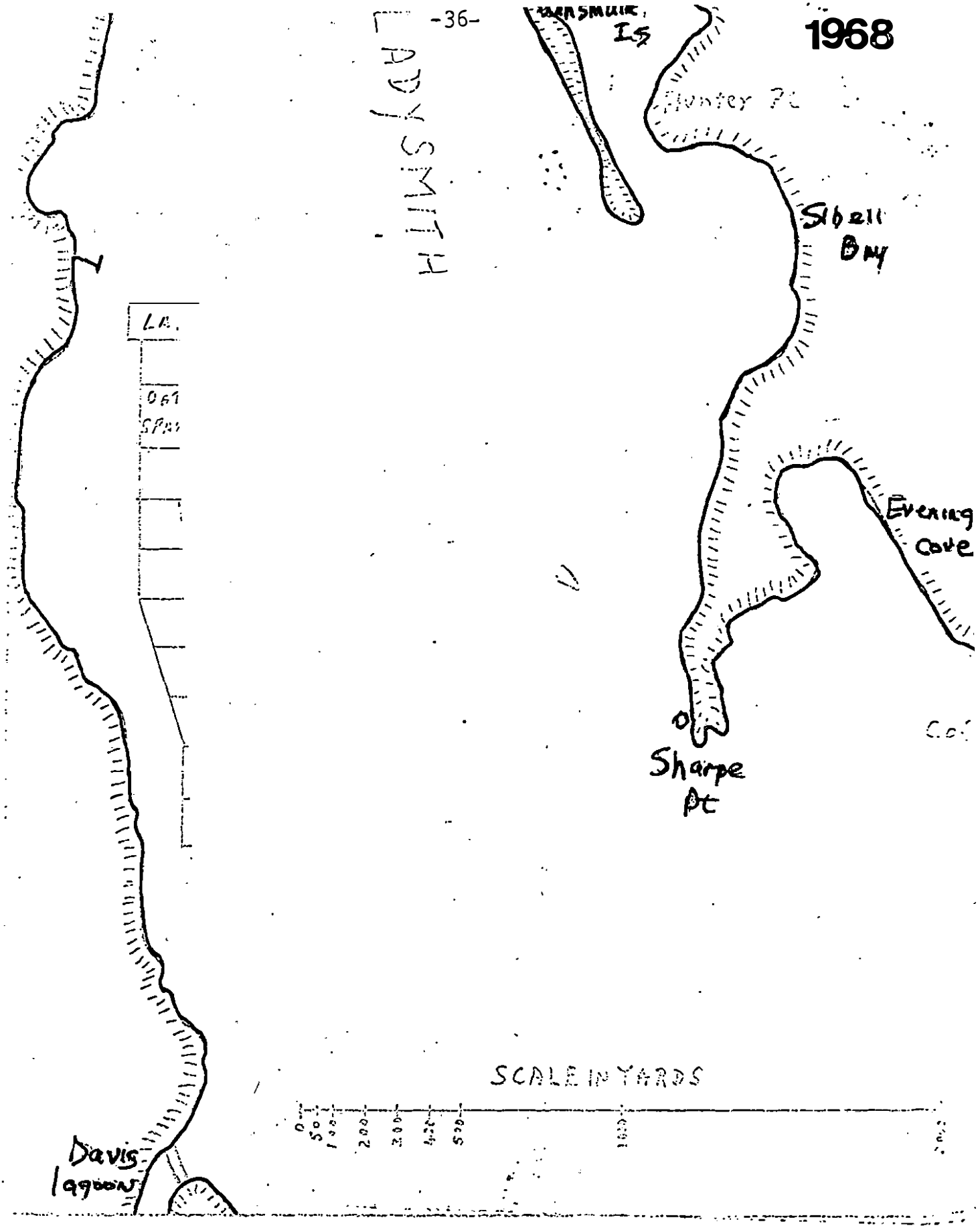
1968



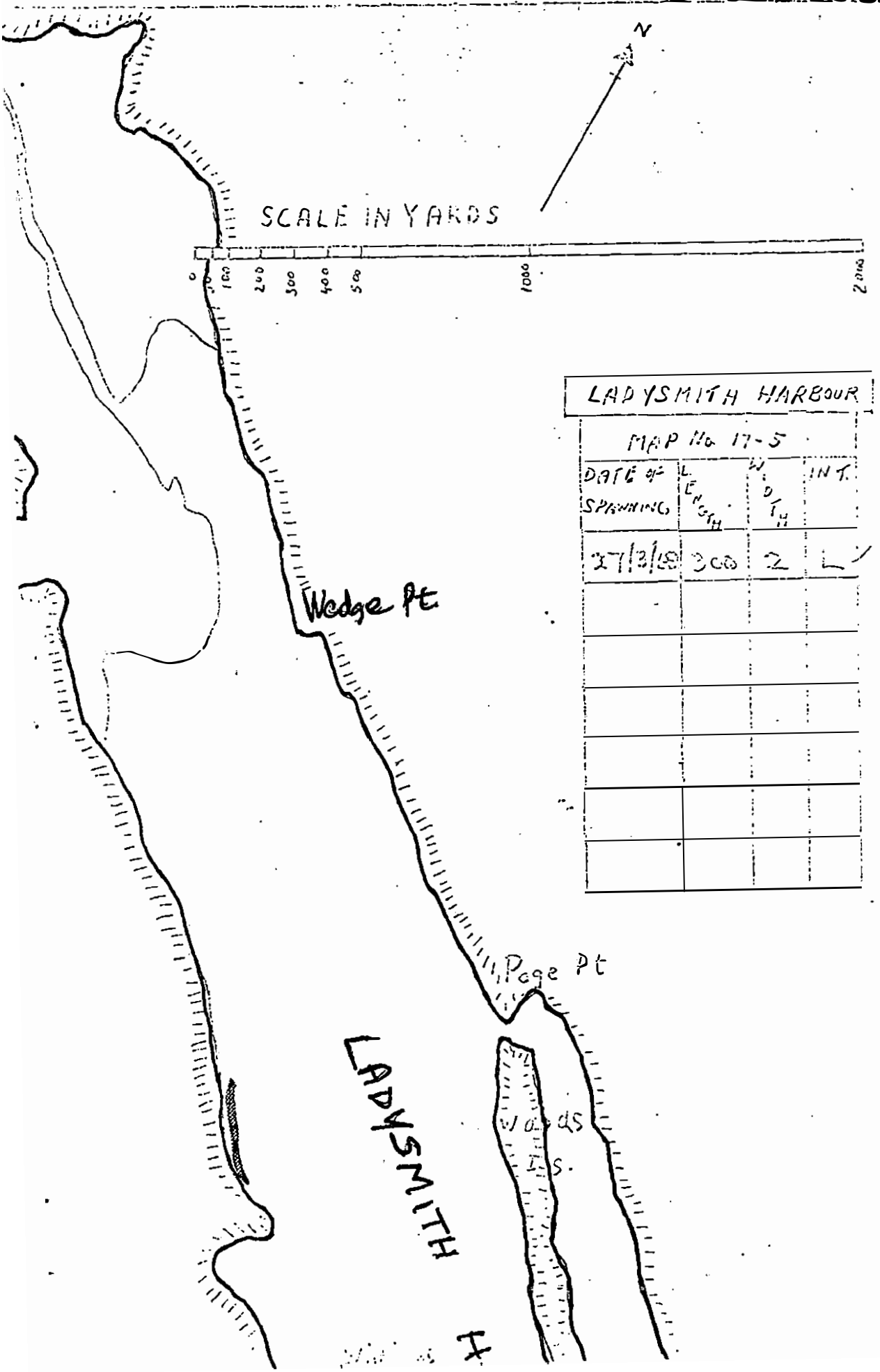
LADYSMITH HARBOUR

MAP No. 17-5

DATE OF SOUNDING	LENGTH	No. OF FIX	INITIALS


1968
Hunter Pt
Shell Bay
Evening Cove
Sharpe Pt
Davis lagoon
SCALE IN YARDS
0
500
1000
2000
3000
4000
5000
10000

1969

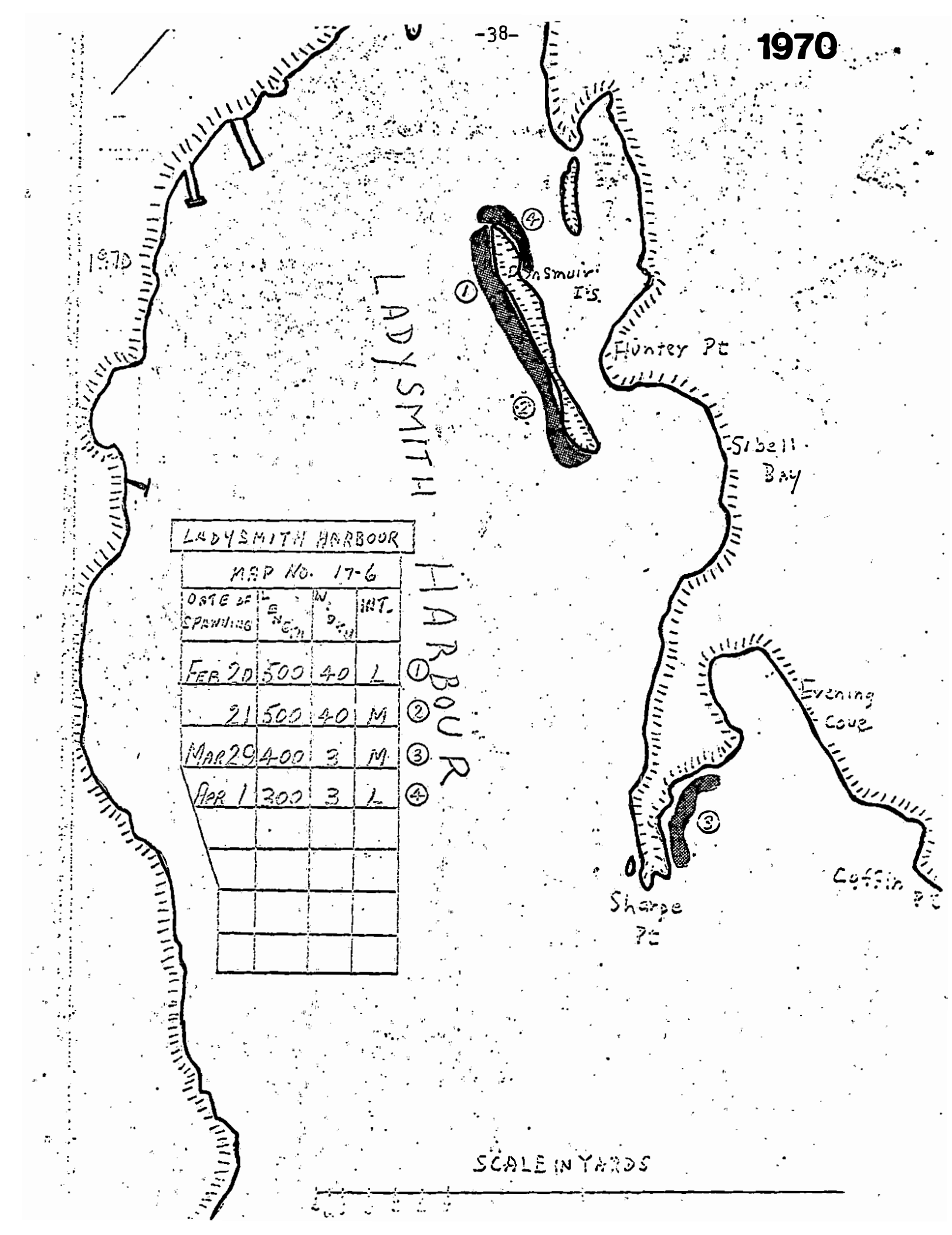


LADYSMITH HARBOUR

MAP No. 17-5

DATE OF SPAWNING	LENGTH	WIDTH	INT.
27/3/69	300	2	L

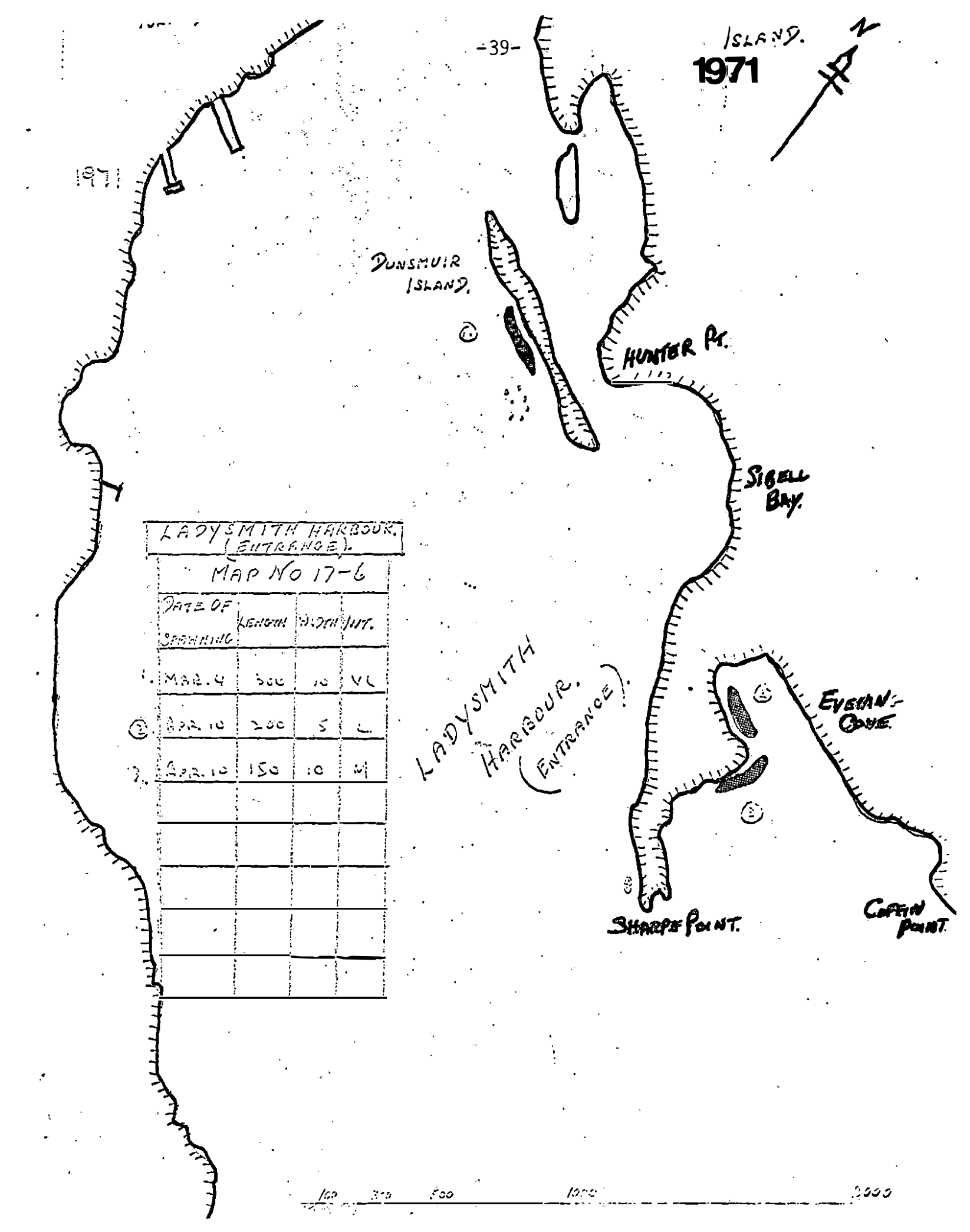
1970



LADYSMITH HARBOUR

MAP NO. 17-6

DATE OF SPAWNING	LENGTH	WIDTH	INT.	
FEB 20	500	40	L	①
21	500	40	M	②
MAR 29	400	3	M	③
APR 1	300	3	L	④

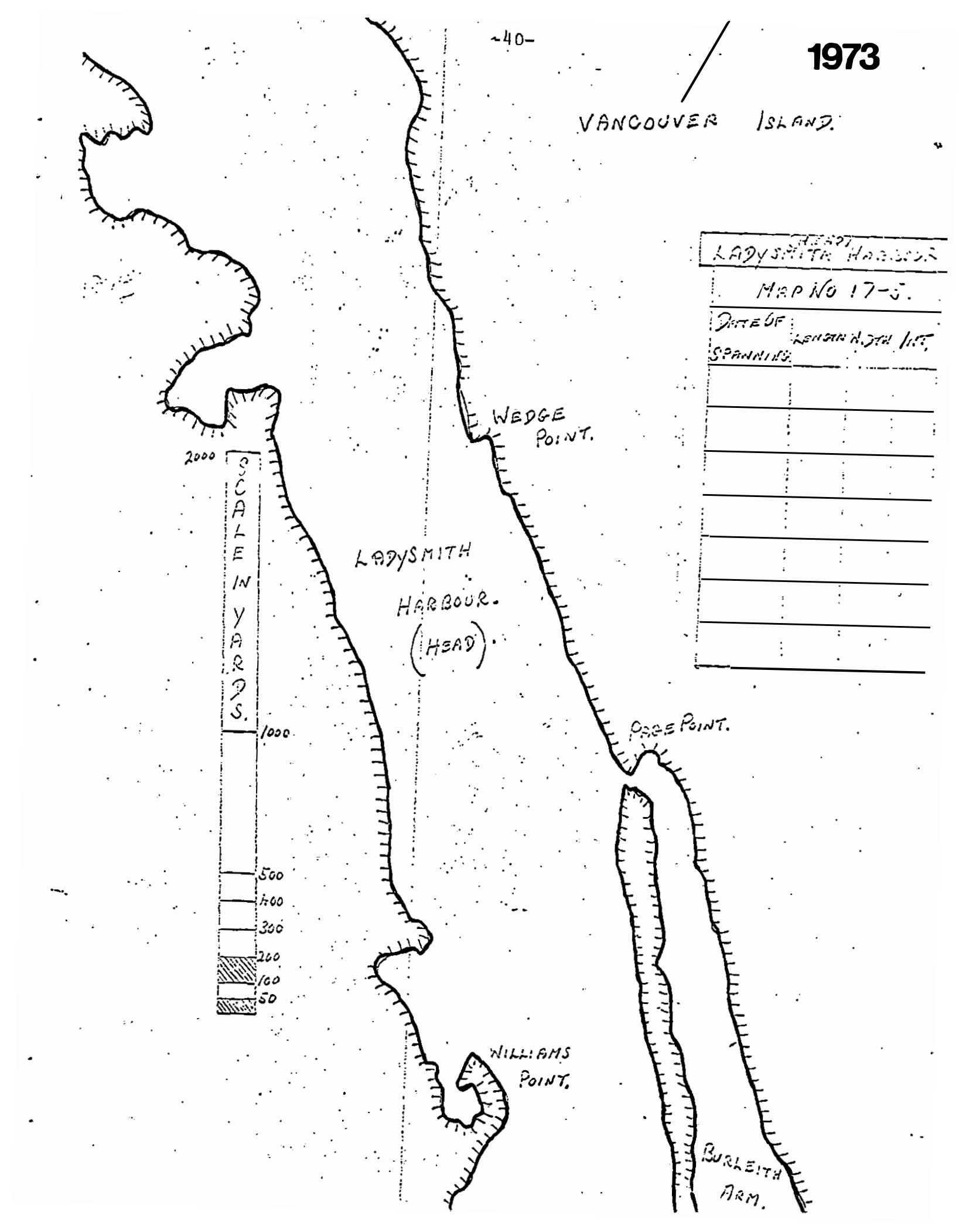


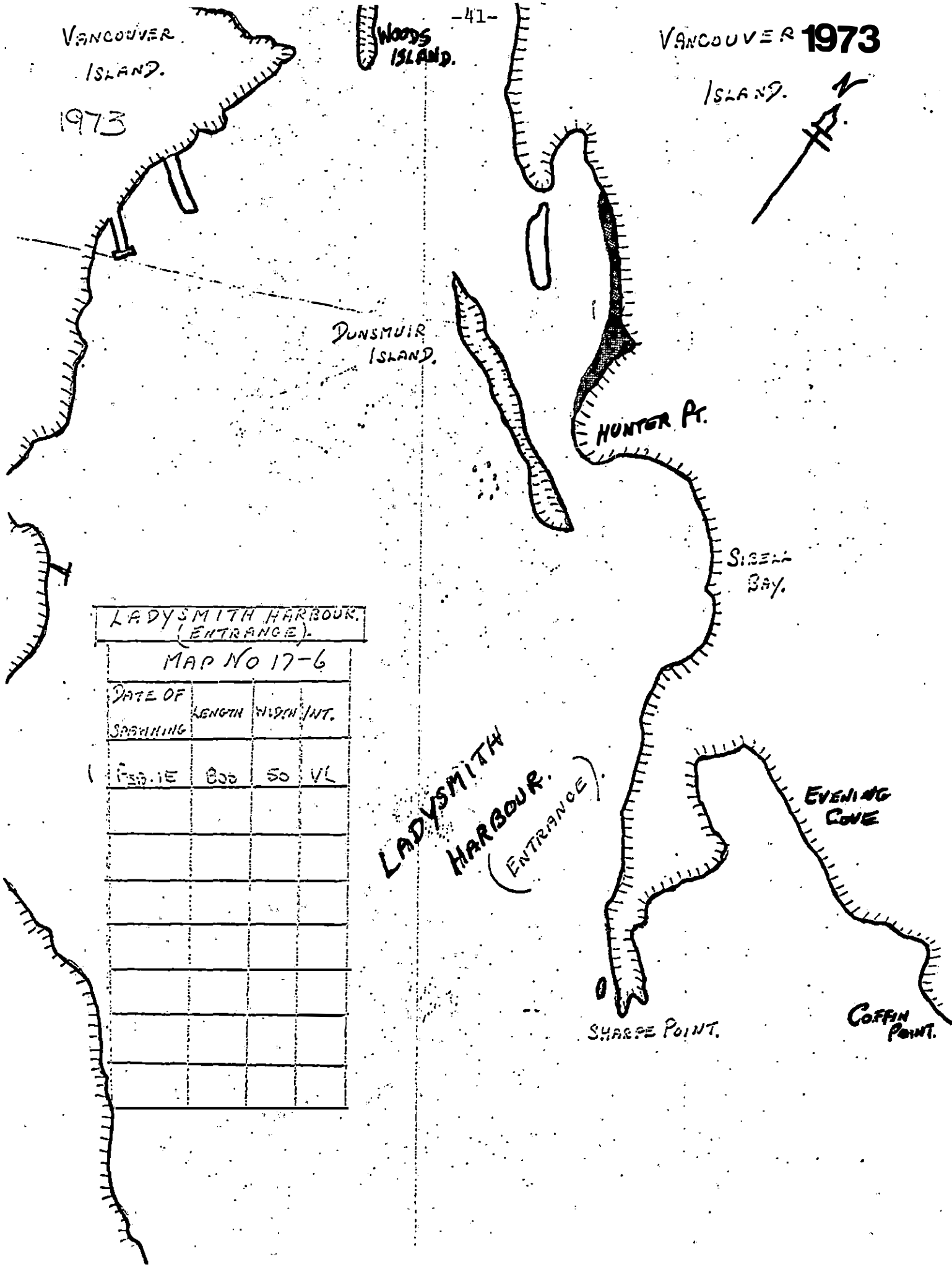
LADYSMITH HARBOUR. (ENTRANCE).

MAP NO 17-6

	DATE OF SPAWNING	LENGTH	WIDTH	INT.
1.	MAR. 4	500	10	VL
2.	APR. 10	200	5	L
3.	APR. 10	150	10	M

1973

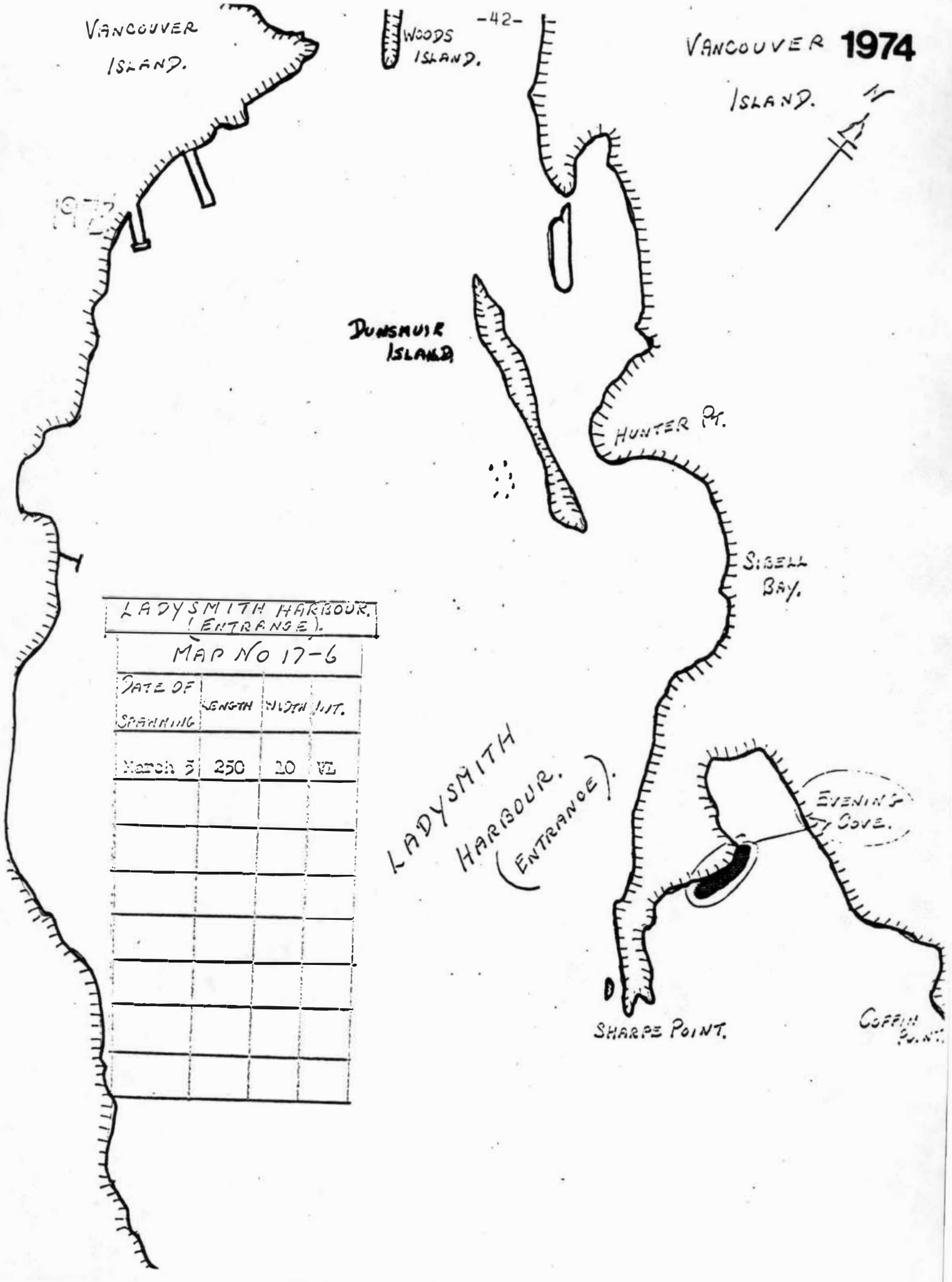

VANCOUVER ISLAND.
LADYSMITH HARBOUR
MAP NO 17-5.
DATE OF SOUNDINGS
LENGTH WIDTH INT.
WEDGE POINT.
LADYSMITH HARBOUR. (HEAD).
PAGE POINT.
WILLIAMS POINT.
BURLEITH ARM.
SCALE IN YARDS.
2000
1000
500
400
300
200
100
50



LADYSMITH HARBOUR. (ENTRANCE).

MAP NO 17-6

DATE OF SOUNDING	LENGTH	WIDTH	INT.
FEB. 1E	800	50	VL



LADYSMITH HARBOUR. (ENTRANCE).

MAP NO 17-6

DATE OF SPANNING	LENGTH	WIDTH	VIT.
March 5	250	20	VL

APPENDIX D

Photographs of intertidal mud flats at uppermost end of Ladysmith Harbour.

PLATE 19-20	Mud flats (intertidal) at head of Ladysmith Harbour. Note log booms.
21-22	Mud flats (intertidal) at head of Ladysmith Harbour. Note debris.
23	Mud flats (intertidal). Note debris.



PLATE NO. 19 MUD FLATS (INTERTIDAL) AT HEAD OF LADYSMITH HARBOUR. SHOWING LOG BOOMS.



20



PLATE NO. 21 MUD FLATS (INTERTIDAL) AT HEAD OF LADYSMITH HARBOUR. NOTE DEBRIS.

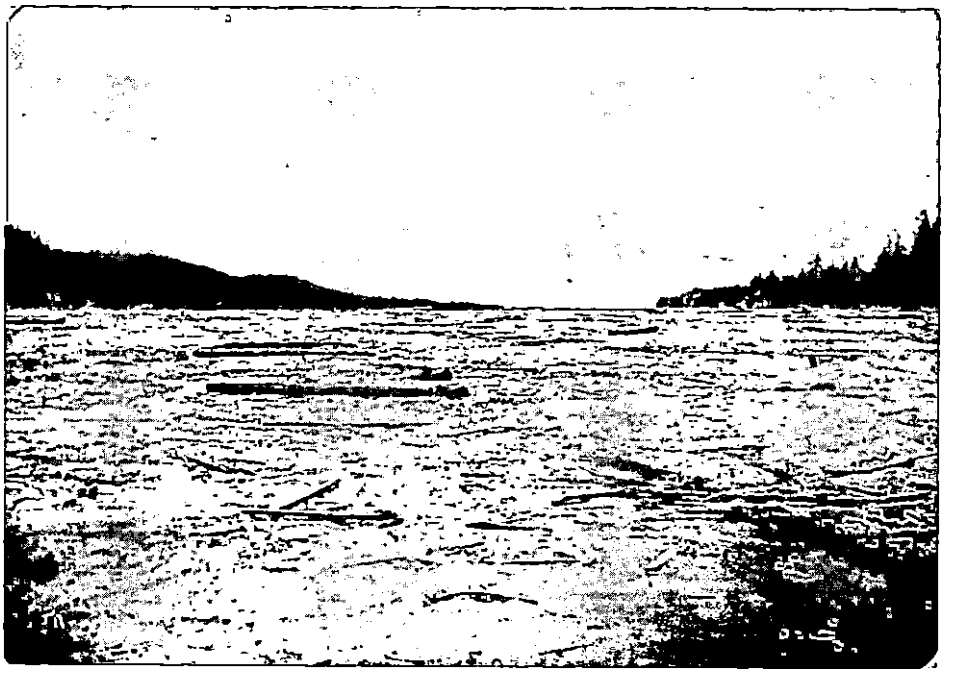




PLATE NO. 23 MUD FLATS (INTERTIDAL) AT HEAD OF LADYSMITH HARBOUR



PLATE NO. 1 RAFT BOOMING OF LOGS IN FRONT OF DOMAN'S INDUSTRIES SAWMILL (LADYSMITH FOREST PRODUCTS)



2 LOG BOOMING GROUNDS ADJACENT IVY GREEN PARK SALTMARSH (CLAY/MUD SUBSTRATE)

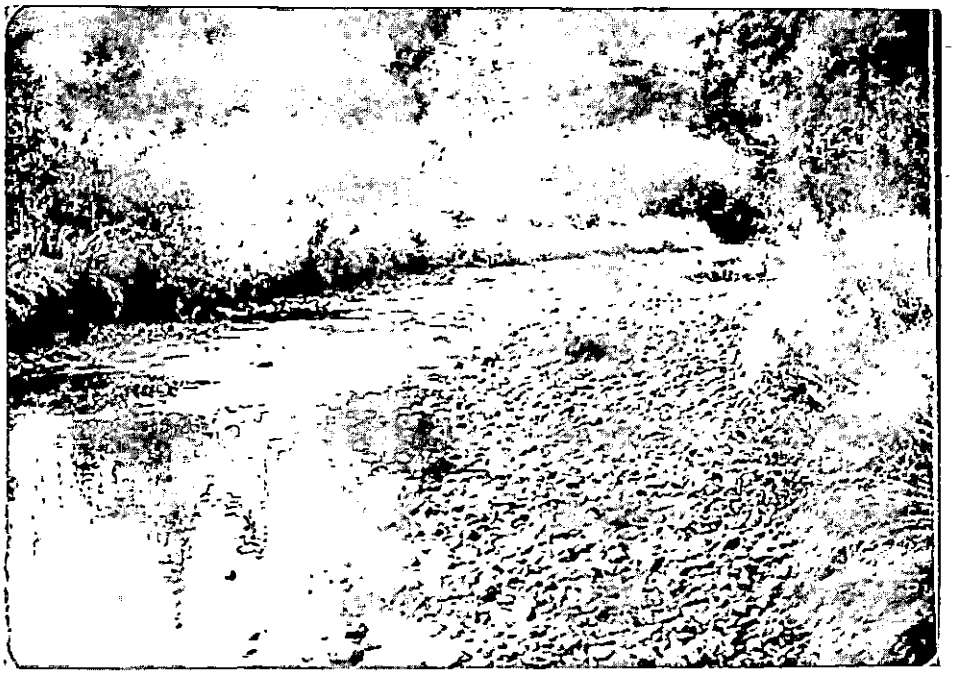
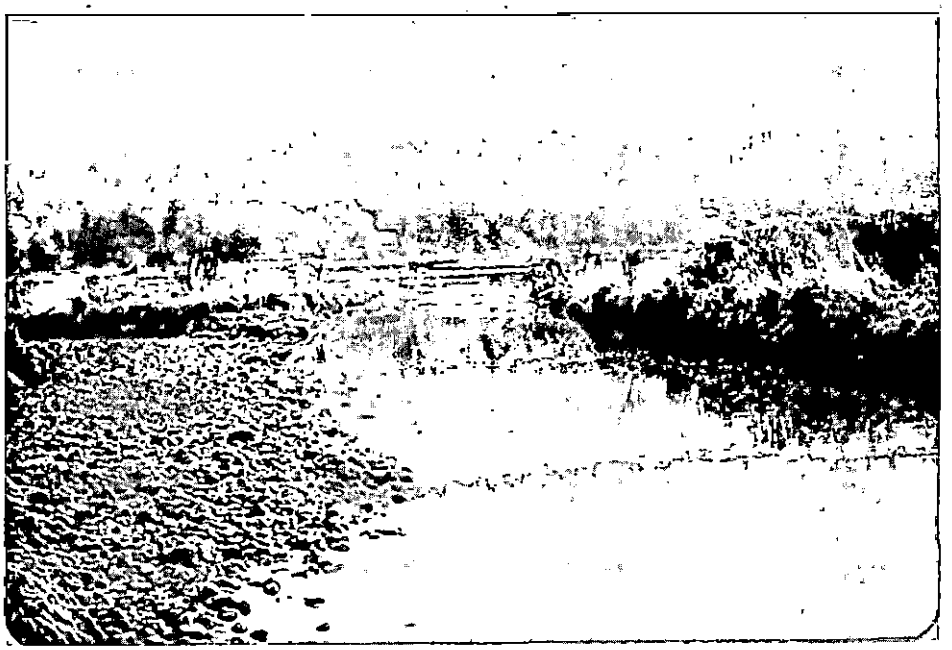


PLATE NO. 3 BUSH CREEK, 150' UPSTREAM FROM ITS MOUTH. SALTMARSH VEGETATION BORDERS THE STREAM COURSE.



4 BUSH CK., LOOKING DOWNSTREAM TO MOUTH.



PLATE NO. 5 BUSH CREEK ASSOCIATED SALTMARSH



6 BUSH CREEK ESTUARINE AREA



PLATE NO. 7 ROCKY CK. STREAMBED BEHIND SALTAIR LOGGING COMPANY PROPERTY



.8 .ROCK CK. NEAR MOUTH, SHOWING DIVERTED CHANNEL AROUND SALTAIR LOGGING CO.



PLATE NO. 9 LADYSMITH HARBOUR, IMMEDIATELY IN FRONT OF ROCKY CK.



PLATE NO. 10 SALTMARSH, IVY GREEN PARK ADJACENT TO MOUTH OF BUSH CK.



11 DAVIS LAGOON, SHOWING SALTMARSH NEAR MOUTH OF STOCKING LAKE CK.



PLATE NO. 12 ESTUARINE AREA WHERE STOCKING LAKE CK. FLOWS INTO STUART CHANNEL



13 DEBRIS AND LOOSE LOGS FROM LOG HANDLING FACILITIES AT HEAD OF LADYSMITH HARBOUR

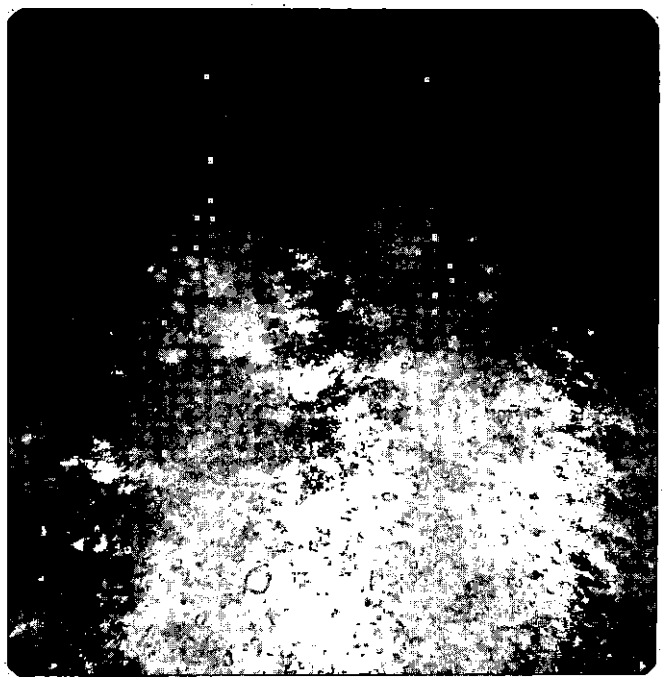
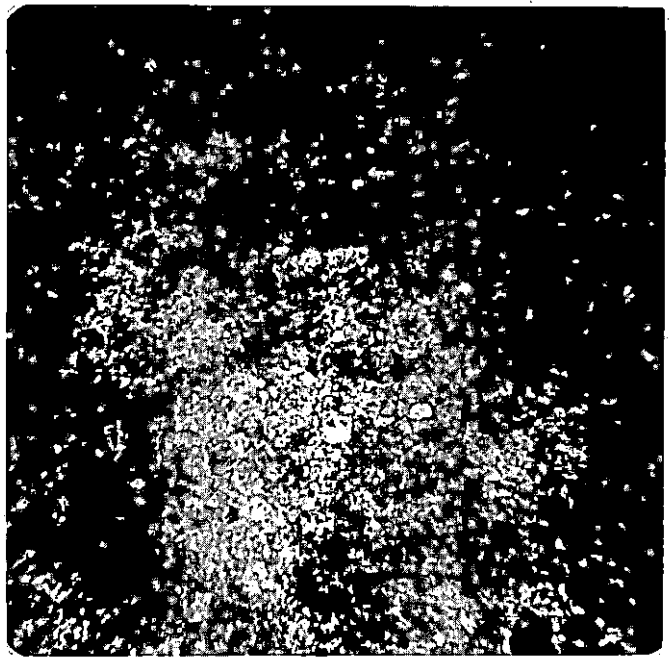


PLATE NO. 14 SUBSTRATE
LOWER INTERTIDAL AREA
BUSH CK. ESTUARY



15 MUD FLATS, DAVIS LAGGON
MOUTH OF STOCKING LAKE CREEK

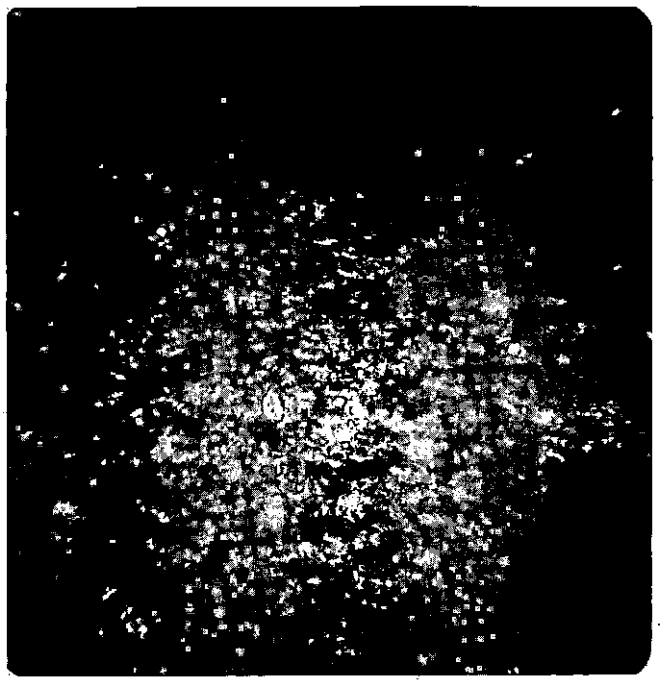


PLATE NO. 16 LOWER INTERTIDAL REGION
HOLLAND CK. ESTUARY



17 SALTMARSH (with some debris)
HOLLAND CR. below supra-
tidal line



PLATE NO. 18 SUBSTRATE SAMPLE SITE 5 (S_5)
INTERTIDAL AREA (see Map)^5